

SANTANU CHAKRABORTY

Staff Software Engineer | Payments Risk & Fraud Platforms | Distributed Systems | Agentic AI

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SUMMARY

Staff Software Engineer with 10+ years architecting global payments, fraud and risk platforms that 1,000+ issuing banks use to approve or decline transactions in real time. Leading distributed systems, observability, full-stack and agentic AI initiatives across global teams.

TECHNICAL SKILLS

- **Backend & Distributed Systems:** Java, Spring Boot, Microservices, REST APIs, Kafka, Redis, Python, Event-Driven Architecture, API Design, Real-Time Systems, Distributed Caching, Performance Engineering
- **AI, MCP & Agentic Engineering:** Agentic AI, Model Context Protocol (MCP), RAG, LLM Applications, Agentic Workflow Automation, Tool Integration, Prompt & Context Engineering
- **Frontend & Full-Stack:** React, TypeScript, Redux, Micro-Frontends, Module Federation, Angular, Frontend Architecture
- **Data, Cloud & Observability:** DB2, SQL, StarRocks, MongoDB, Docker, Kubernetes, Jenkins, CI/CD, AWS, Prometheus, Splunk, SLA/OLA Monitoring, HashiCorp Vault, Git, Jira, Maven, Release Engineering
- **Security & Testing:** SAST, DevSecOps, OWASP Top 10, Secure SDLC, JUnit, Mockito, Jest, Selenium, Cypress, JMeter, Pen Testing

PROFESSIONAL EXPERIENCE

Visa Inc. | Staff Software Engineer

May 2022 – Present, USA

Lead technical initiatives for Visa Risk Manager (VRM), an ~\$85M/year fraud-decisioning platform serving 1,000+ issuing banks.

- **Fraud Reporting & Analytics Suite:** Architected VRM's customer-facing suite, cutting analytics support tickets by **62%**. Built on React and Spring Boot microservices over pre-aggregated stores that isolate reporting traffic off the latency-critical authorization path.
- **Rule Management Platform Modernization:** Led the rebuild for real-time decisioning workflows, cutting rule lookup from **8 mins to 12 secs** across **800+ active rules**. Defined the end-to-end architecture from Spring Boot microservices through React micro-frontends.
- **SLA Observability Platform:** Architected coverage across **8.34M+ monthly requests** on production endpoints, cutting SLA-miss detection from **3 weeks to under a day** and avoiding ~\$400K in annual contractual SLA credits through data drilldowns.
- **Agentic AI Security Remediation:** Closed **130 of 130** security findings, cutting per-finding remediation from **a day to 45 minutes**. The framework, built on compounding-engineering principles, produced **19 patterns and 5 workflows** now standard across **40+ repositories**.
- **Risk-Analyst Workflow Automation:** Partnered with 14 global risk analysts to automate KPI calculation, testing, validation and client-ready reporting, cutting per-rule processing from **2 hours to 4 minutes** and saving **~9,000 analyst-hours a year (~\$1.1M)**.
- **Enterprise MCP Server:** Built a Model Context Protocol server exposing rule, KPI and reporting data as permission-scoped tools, letting AI agents act on governed production fraud data inside enterprise guardrails — adopted by **6 teams**.

Visa Inc. | Senior Software Engineer

Jul 2016 – May 2022, USA

- **VRM Long-Term Architecture:** Designed **50+ REST APIs** for the re-architecture of Visa Risk Manager and set the API conventions the team still builds against.
- **React Micro-Frontend Migration:** Migrated **85+ Angular pages** to React and TypeScript micro-frontends on Module Federation with **zero production incidents**, improving feature lead time by **35%** while preserving years of embedded fraud business logic.
- **Rule Authoring & Operational Analytics:** Developed features for rule authoring, testing, and approval workflows that decide whether card transactions are approved or declined, and cut diagnosis of authorization delays from **6 hours to 40 minutes** with Splunk dashboards for VisaNet.

Walmart Global Tech (@WalmartLabs) | Software Engineer Intern

Jun 2015 – Dec 2015, USA

- **Last-Mile Grocery Delivery:** Built the Android client and REST services for one of Walmart's early driver-facing apps, enabling shipment loading, real-time tracking, and delivery validation on enterprise handhelds.

Infosys Ltd. | Senior Systems Engineer

Sep 2011 – Jul 2014, India

- **Skype on XFINITY (Comcast) & EA Origin:** Built set-top-box video-calling features — audio/video, messaging and file transfer via the Skype API — and drove full-stack feature development and automation for the EA Sports' Origin gaming platform.

RECOGNITION & PUBLICATIONS

- **Guest Lecturer, Stanford University (2x, 2025–2026)** — selected to represent Visa in a Visa–Stanford collab on payments fraud and risk.
- **Best Financial Crime Prevention Solution, Pay360 Awards 2023** — awarded to the Visa Risk Manager team for issuer fraud prevention.
- **TD Commons Publication, 2021** — co-authored “Monitoring Fraudulent Transactions at Merchant Level in Real Time Payment.”

EDUCATION

M.S., Software Engineering	San Jose State University, USA	2016
B.E., Computer Science & Engineering	Visvesvaraya Technological University, India	2011
Certificate, Information Technology Project Management	Stanford University, USA	2022